InsightPilot AI — System README & Technical Manual

Enterprise Decision Intelligence, Deterministic Root-Cause Attribution & Cryptographic Evidence Lineage

1. Executive Summary & Problem Context

InsightPilot AI is an enterprise decision intelligence platform that automates the transition from descriptive anomaly detection to prescriptive operational recovery. Traditional BI tools indicate *what* occurred (e.g., -$1.23M revenue deficit), but investigating *why* requires weeks of cross-silo dataset triage across ERP, CRM, and EDI systems. InsightPilot AI bridges this gap using deterministic analytics engines, LangGraph 11-node orchestration, and SHA-256 cryptographic provenance.

Core Architectural Principle: *'Deterministic systems own quantitative truth. LangGraph orchestrates investigation. AI explains grounded facts.' LLMs are strictly forbidden from calculating metrics or inventing evidence.*

2. The 7-Screen Executive Analytical Journey

Screen / Route	Primary Capability	Enterprise Impact
1. Command Center (/)	Real-time revenue anomaly triage & data source synchronization	Detects -$1.23M deficit (-7.97%) across 8 synced enterprise systems.
2. Root Cause (/root-cause)	Deterministic waterfall attribution & 6-milestone timeline	Attributes 43.2% ($550K) to Atlanta DC stockout; 100% variance explai
3. Investigation (/investigation)	LangGraph 11-node state graph trace & agent replay	Full agentic audit trail with tool execution logs & confidence scores.
4. Decision Graph (/decision-graph)	Interactive 6-column DAG mapping Anomaly -> Actions	Visual causality pipeline with stage stepper & node inspector.
5. Evidence Explorer (/evidence)	12 multi-modal records with SHA-256 verification	Cryptographic provenance proof for ERP, EDI, CRM, and POS data.
6. Recommendations (/recommendations)	2x2 Prioritization Matrix & What-If Elasticity Sandbox	Priority 1 Stock Transfer ($484K recovery); $757.6K modeled recovery
7. Executive Briefing (/briefing)	Role-tailored synthesis (CFO, VP Supply Chain, COO, Sales)	Audit-ready governance briefings with 1-click boardroom PDF export.

3. Technology Stack & Verification

Layer	Technologies	Verification Level
Backend API	Python 3.11+, FastAPI (ASGI), Pydantic v2, Uvicorn	305/305 Unit/Integration Tests Passing
Deterministic Core	Pure Python Engines (KPI, Driver, Evidence, Confidence)	100.000% Mathematical Precision ($1.23M locked)
AI Orchestration	LangGraph 11-Node StateGraph, Groq LLaMA 3.3 70B, Gemini 2.5	Grounded facts only; calibrated abstention gate
Frontend Web App	Next.js 14.2 (App Router), React 18, Tailwind CSS, Lucide	10/10 Static Routes Pre-rendered Cleanly
Lineage & Security	8 Normalized CSV Schemas (43K+ rows), SHA-256 Hashes	100% Cryptographic Digest Verification

4. Quick Start & Execution Commands

Backend API Server:

```
pip install -r requirements.txt
uvicorn backend.app.main:app --host 127.0.0.1 --port 8000
```

Frontend Web Application:

```
cd frontend/next-app && npm install && npm run dev
# App runs at http://localhost:3000
```

Verification Suite:

```
python tests/validate_dataset.py
python -m unittest discover -s tests -t . -p 'test_*.py'
cd frontend/next-app && npm run build
```